The Hölder and Minkowsk Inequalities

Lemma A.1: (Hölder's inequality) for $x, y > 0$ s.t. $x+y=1$ and (possibly complex sequences) $\{a_i, i=1,2,\dots,n\}$ and $\{b_i, i=1,2,\dots,n\}$.

$$\left|\sum_{i=1}^{n} a_i b_i\right| \leq \left(\sum_{i=1}^{n} |a_i|^{1/x}\right)^{x} \left(\sum_{i=1}^{n} |b_i|^{1/y}\right)^{y}$$

If $x = y = 1/2$ then we have the Cauchy-Schwarz inequality.

Lemma A.2: (Minkowski inequality) For positive numbers A_i and B_i, $i = 1, 2, \dots, n$

$$\left(\sum_{i=1}^{n} (A_i + B_i)^p\right)^{1/p} \leq \left(\sum_{i=1}^{n} A_i^p\right)^{1/p} \left(\sum_{i=1}^{n} B_i^p\right)^{1/p}$$

with an equality iff

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}$$

These inequalities are also true with integrals. Just change $\sum$ with $\int$ on interval $[a, b]$

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